

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Electrical Engineering	IIT Gandhinagar	2024-Present	7.7
Class XII	Physics, Chemistry, Maths	SES Junior College, Solapur	2023-2024	85.17
Class X		Damani Vidya Mandir	2021-2022	97.20

- PROJECTS
- Bazaar IITGN: College Marketplace Web Application

[April '26]

(Full-Stack Web Development Hackathon) | Project Link

Built and deployed a production-grade full-stack marketplace platform for the IIT Gandhinagar community using **Next.js**, **FastAPI**, and **MongoDB**, enabling students to buy, sell, and manage pre-owned items with features including product listings, offers, watchlists, reviews, notifications, and an **admin dashboard**.

Developed modular **REST APIs** using **FastAPI** and **MongoDB**; integrated secure **Firebase Authentication** with access restricted to verified IIT Gandhinagar email accounts, ensuring seamless frontend-backend communication.

Enabled real-time buyer-seller communication using **WebSockets**, managed media uploads through **Cloudinary**, and deployed the application with a decoupled architecture using **Vercel**, **Render**, and **MongoDB Atlas** for scalable and reliable hosting.

• Rubik's Cube Solver using C++ & OOP

[May '26 - June '26]

Project Link

Developed a modular **3x3 Rubik's Cube Solver** in **C++** using Object-Oriented Programming with interchangeable cube representations including **3D Array**, **1D Array**, and **Bitboard**.

Implemented multiple graph-search algorithms including **DFS**, **BFS**, **IDDFS**, and **IDA***; accelerated solution search using a **Pattern Database** heuristic for efficient state-space exploration.

Designed an extensible solver architecture using **templates**, **inheritance**, and **polymorphism**, enabling reusable implementations across cube representations and solving strategies.

• ParkEase: Smart Parking Mobile Application

[October '25 - November '25]

(Prof Udit Bhatia, IIT Gandhinagar) | Project Link

Developed a smart parking application using **React Native**, **Expo Router**, and **TypeScript**, enabling users to discover nearby parking spaces through an interactive mobile interface.

Integrated **React Native Maps**, **Expo Location**, and location-based search to provide real-time map visualization, parking spot discovery, navigation, and filtering based on parking type.

Designed a modular application architecture with reusable components, persistent recent-search history using **AsyncStorage**, and responsive navigation to deliver a scalable and user-friendly mobile experience.

• Brain Tumor Classification using Deep Learning

[March '26 - April '26]

(Prof. Himanshu Shekhar, IIT Gandhinagar) | Project Link

Developed an end-to-end deep learning pipeline for **brain tumor classification** using transfer learning with **EfficientNetB0**, **DenseNet121**, and **ResNet50** on MRI images.

Improved classification performance through **CLAHE preprocessing**, denoising, advanced data augmentation, automatic class-weight balancing, and **5-fold Stratified Cross Validation**; combined predictions using a weighted ensemble of CNN models.

Built a comprehensive evaluation framework using confusion matrices, ROC curves, classification reports, and **Grad-CAM** visualizations, with modular training pipelines utilizing Early Stopping, learning-rate scheduling, and model checkpointing.
- TECHNICAL SKILLS

• Programming Languages: C, C++, Python, JavaScript, SQL

• Tools: Git, GitHub, VS Code, Postman, MongoDB, Vercel

• Libraries: React Native, Expo Router, Next.js, FastAPI, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib
- ACHIEVEMENTS

• Codeforces | Rank: Pupil | Highest Rating: 1397

• LeetCode | Contest Rating: 1622 | Solved 400+ Problems

• Secured 99.59 percentile in the MHT-CET 2024 examination among over 3.79 lakh candidates.
- POSITIONS OF RESPONSIBILITY

• Senior Team Member, Blithchron, IIT Gandhinagar

[Aug '25 - Present]

Managed event logistics and collaborated with organizing teams to ensure the smooth execution of institute-wide events.

Coordinated with invited judges and supported registrations, participant coordination, and on-ground event operations across multiple events.